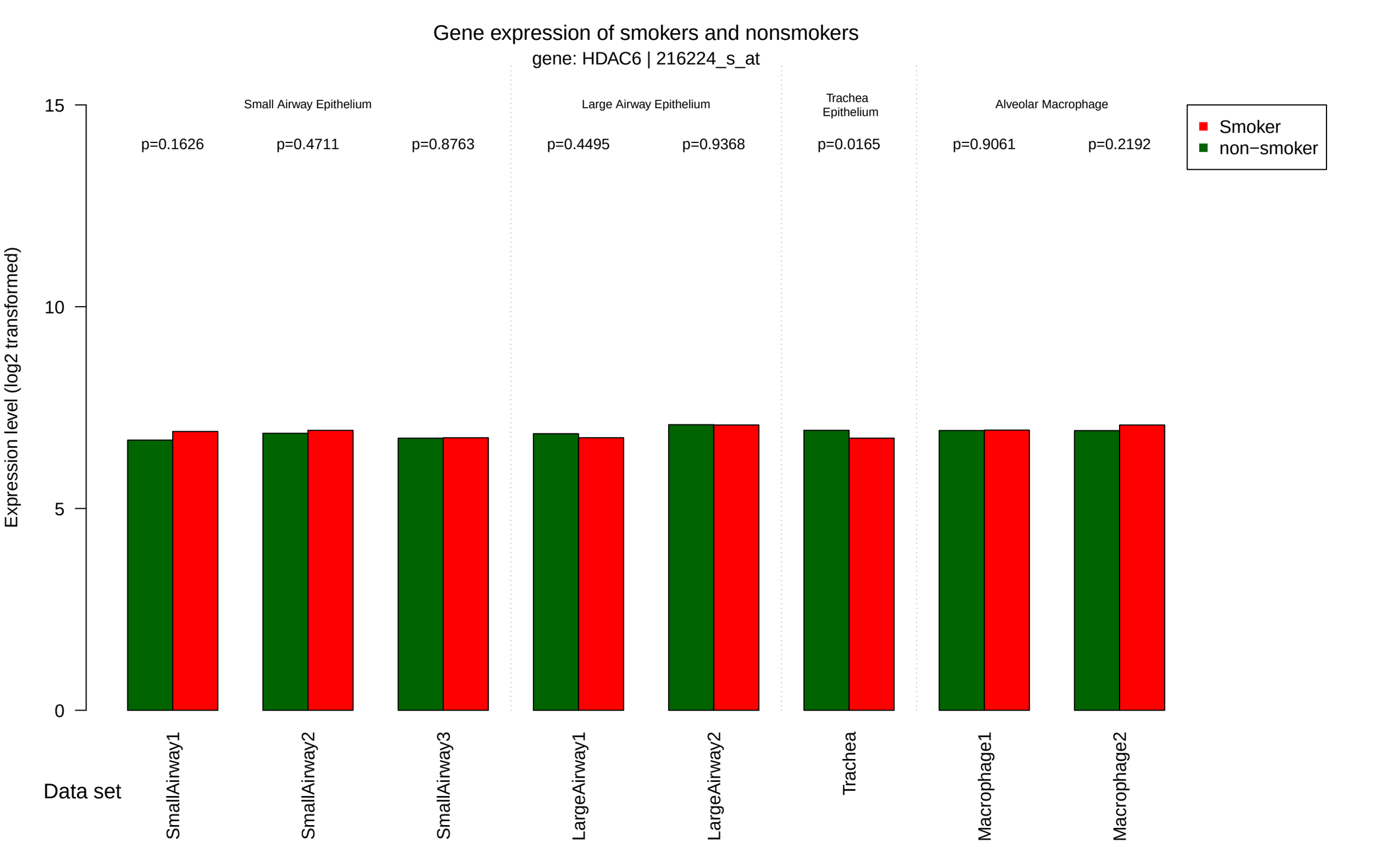
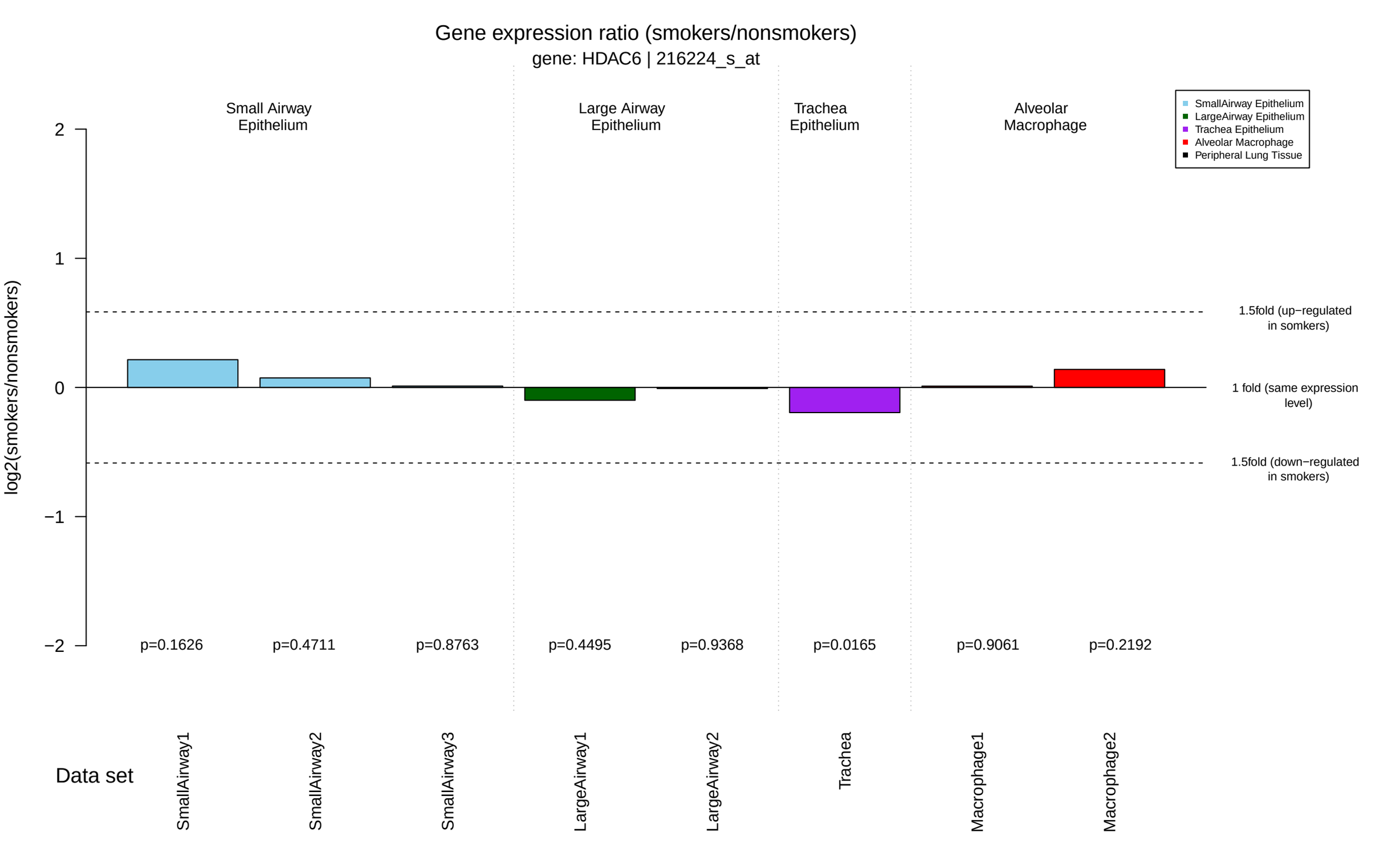
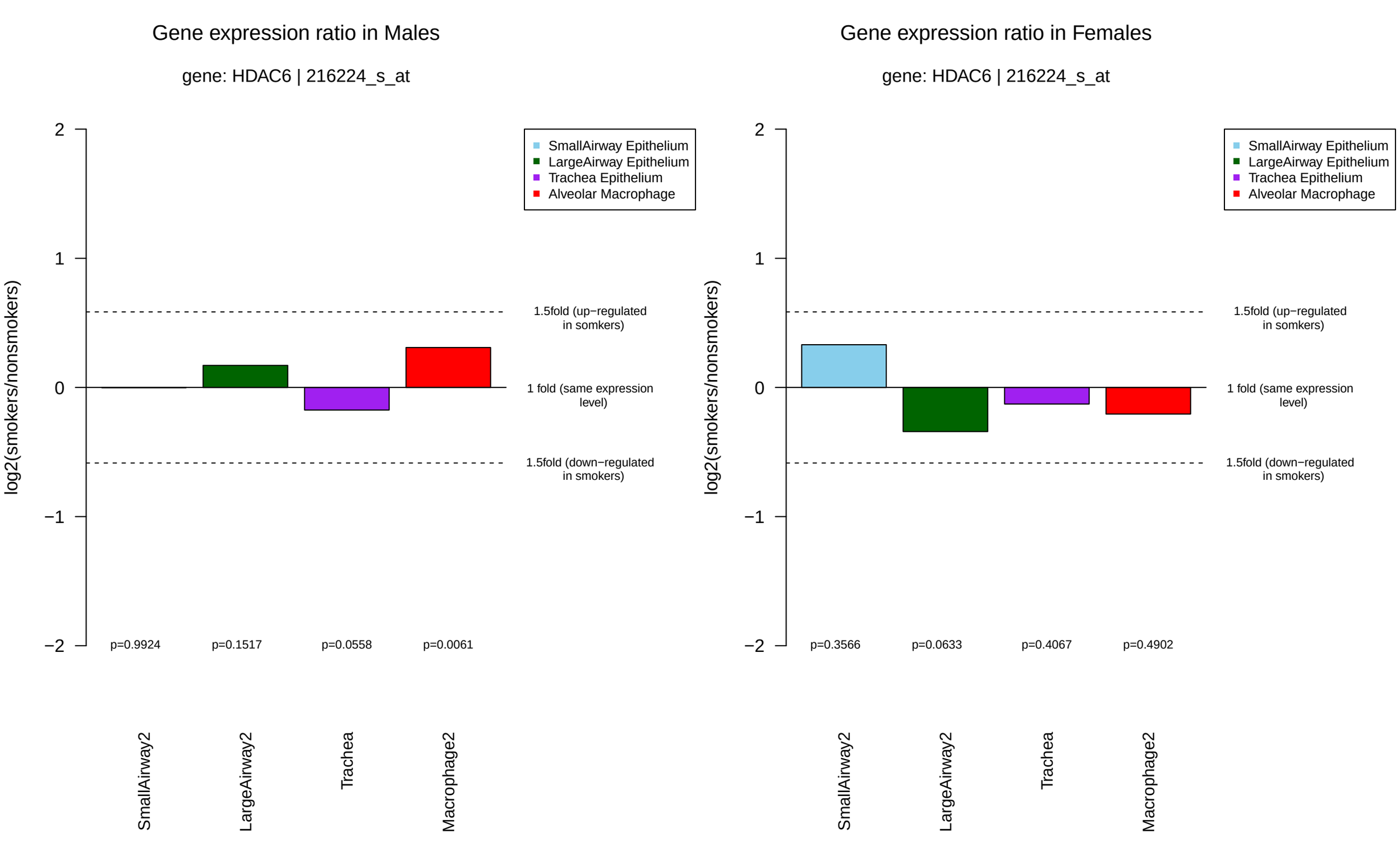
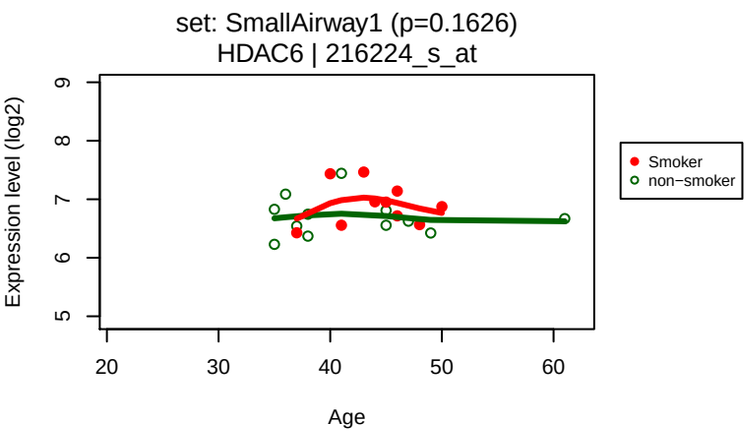
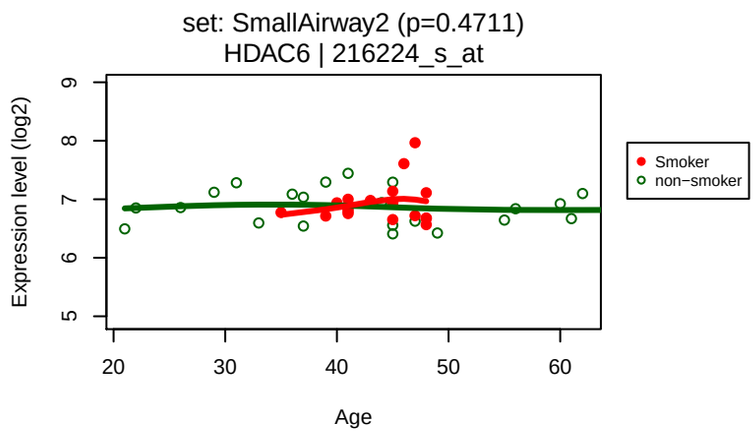
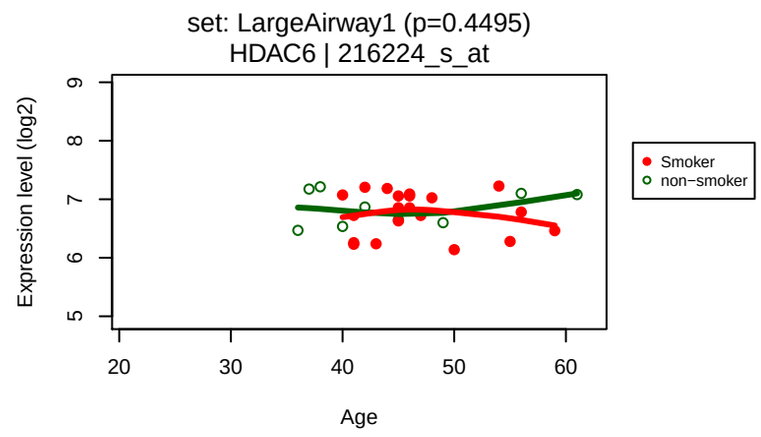

Gene expression of smokers and nonsmokers
gene: HDAC6 | 216224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1626
p=0.4711
p=0.8763
p=0.4495
p=0.9368
p=0.0165
p=0.9061
p=0.2192
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

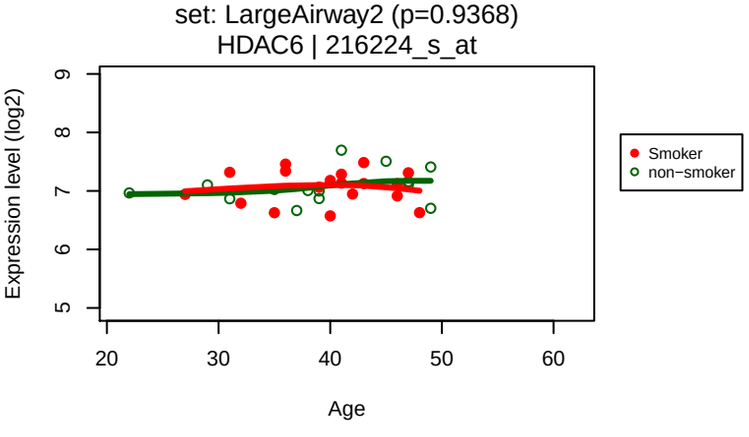

Gene expression ratio (smokers/nonsmokers)
gene: HDAC6 | 216224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1626
p=0.4711
p=0.8763
p=0.4495
p=0.9368
p=0.0165
p=0.9061
p=0.2192
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

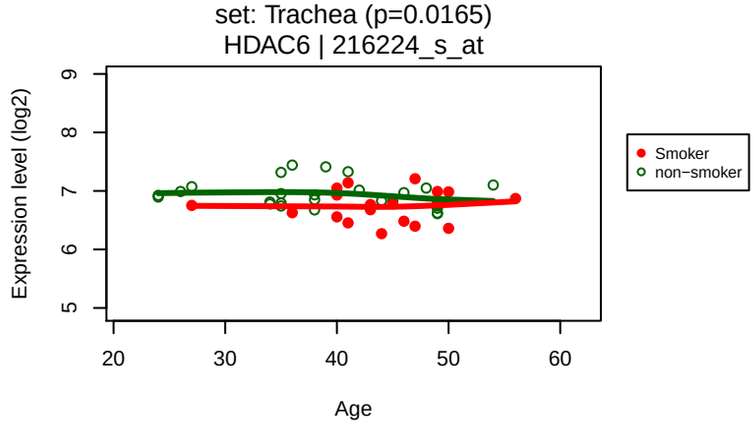

Gene expression ratio in Males
gene: HDAC6 | 216224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9924
p=0.1517
p=0.0558
p=0.0061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC6 | 216224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3566
p=0.0633
p=0.4067
p=0.4902
SmallAirway2
LargeAirway2
Trachea
Macrophage2

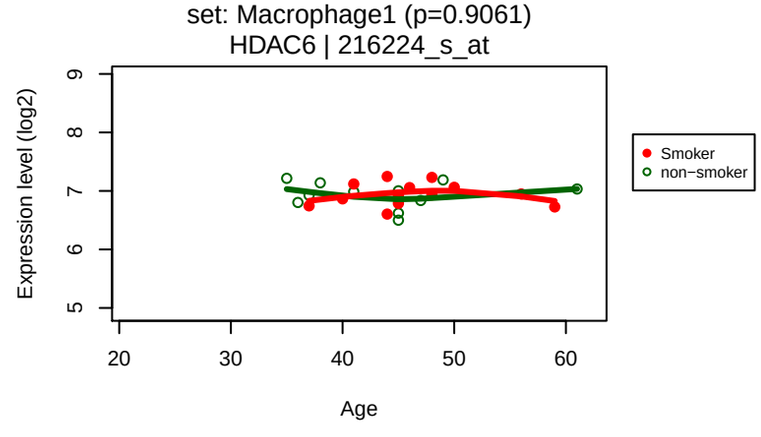

set: SmallAirway1 (p=0.1626)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker

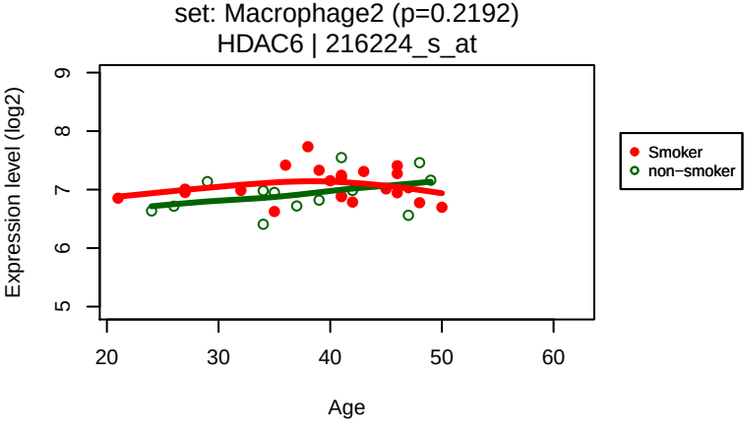

set: SmallAirway2 (p=0.4711)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker

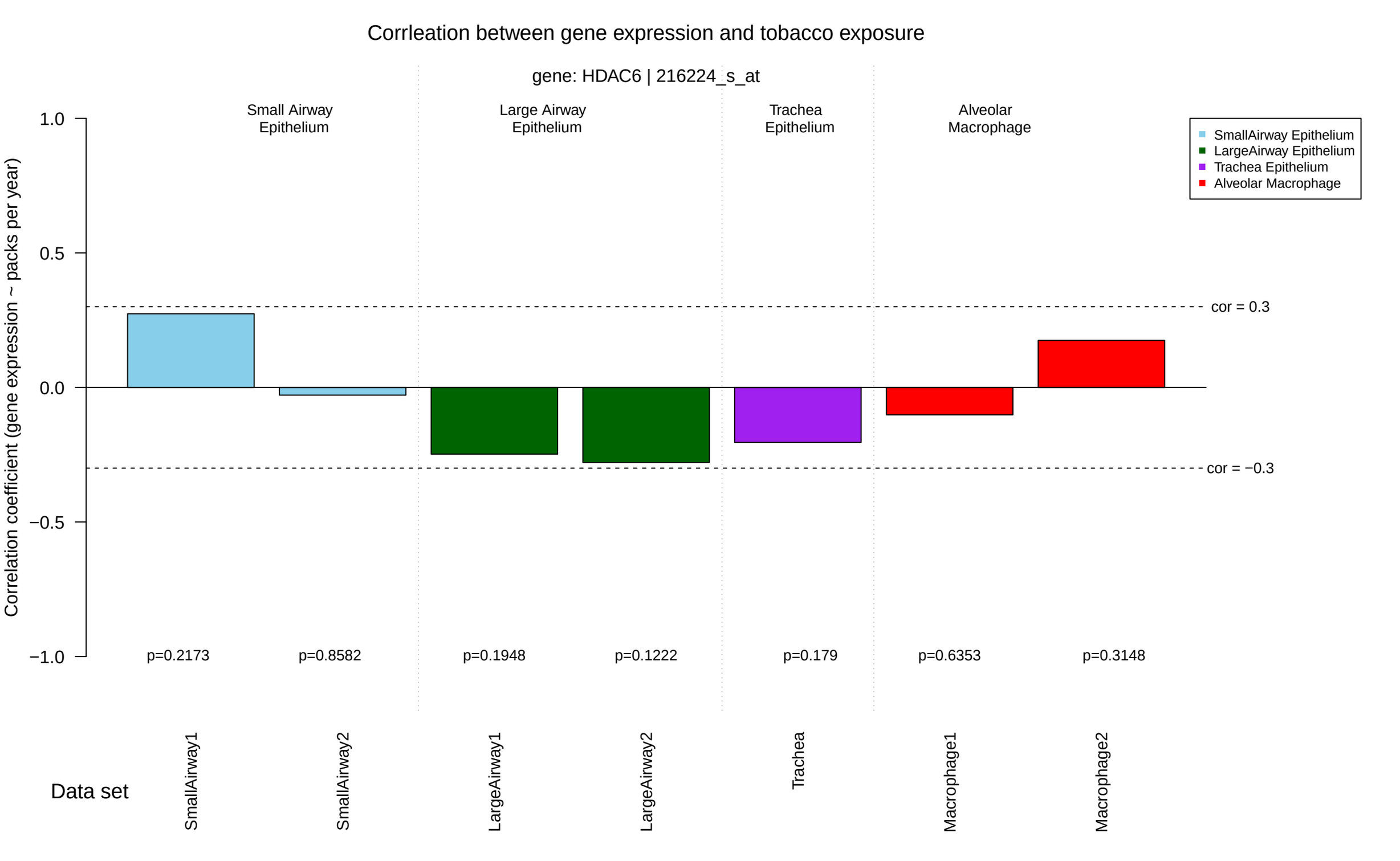

set: LargeAirway1 (p=0.4495)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9368)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0165)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9061)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2192)
HDAC6 | 216224_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC6 | 216224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2173
p=0.8582
p=0.1948
p=0.1222
p=0.179
p=0.6353
p=0.3148
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2